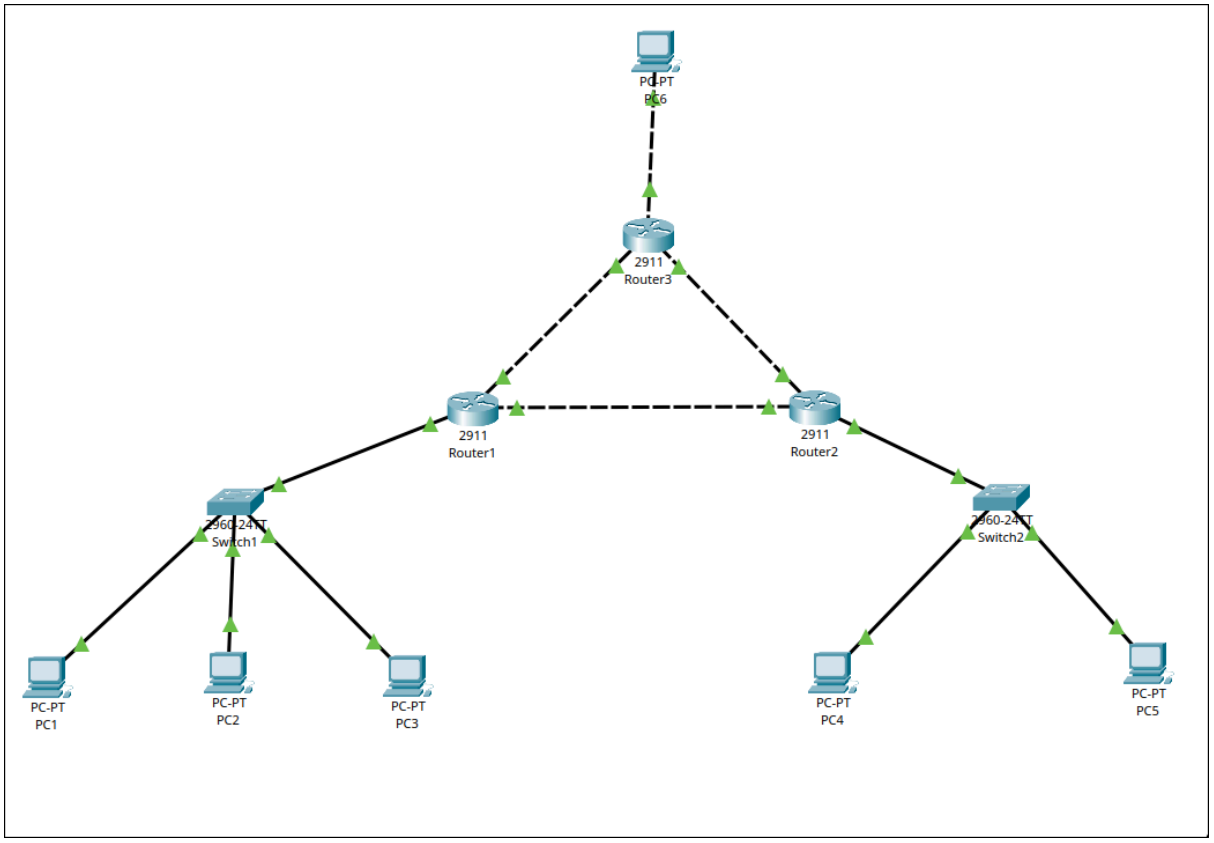
ASSIGNMENT 09
COMPUTER NETWORK
ID: 12341550

PART A: Topology Design



- **WAN:** Three 2911 Routers (Router1, Router2, and Router3) are connected in a full-mesh topology. This design ensures high availability; if one router link fails, traffic can be dynamically or statically rerouted through the third router. These connections utilize Gigabit links.

- **LAN:** To serve the end devices, Router1 and Router2 are connected to 2960 Switches (Switch1 and Switch2, respectively) using Copper Straight-Through cables, which are the standard for connecting devices operating at different OSI layers. Router3 is connected directly to a single end device (PC6).

- **End Devices:** Six PCs are distributed across the network. Switch1 provides connectivity for PC1, PC2, and PC3, establishing the first LAN. Switch2 connects PC4 and PC5, establishing the second LAN.

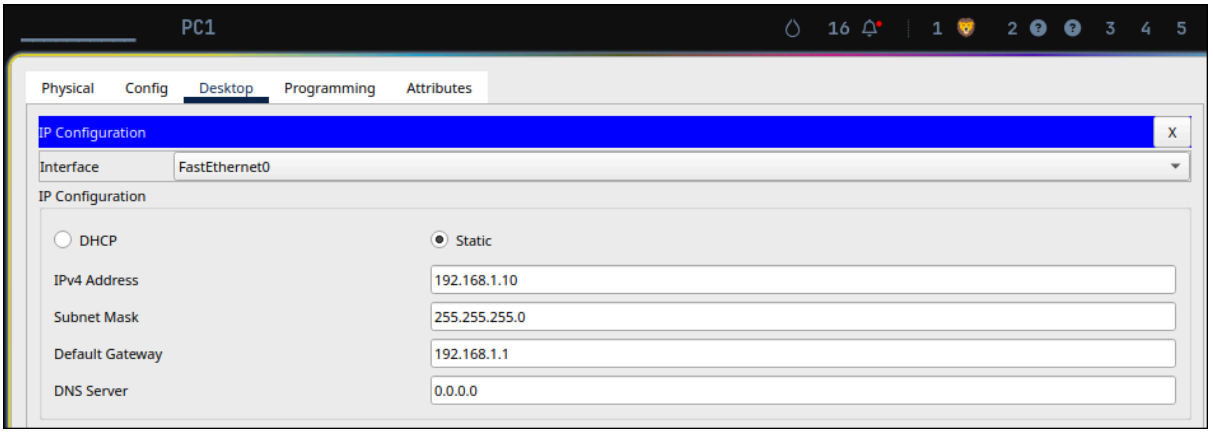
PART B: Configuration

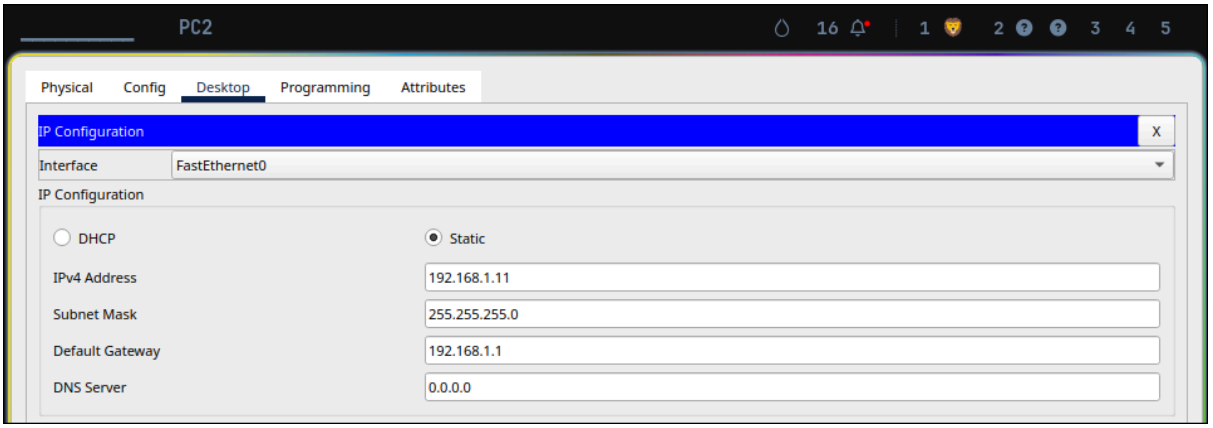
PC configuration:

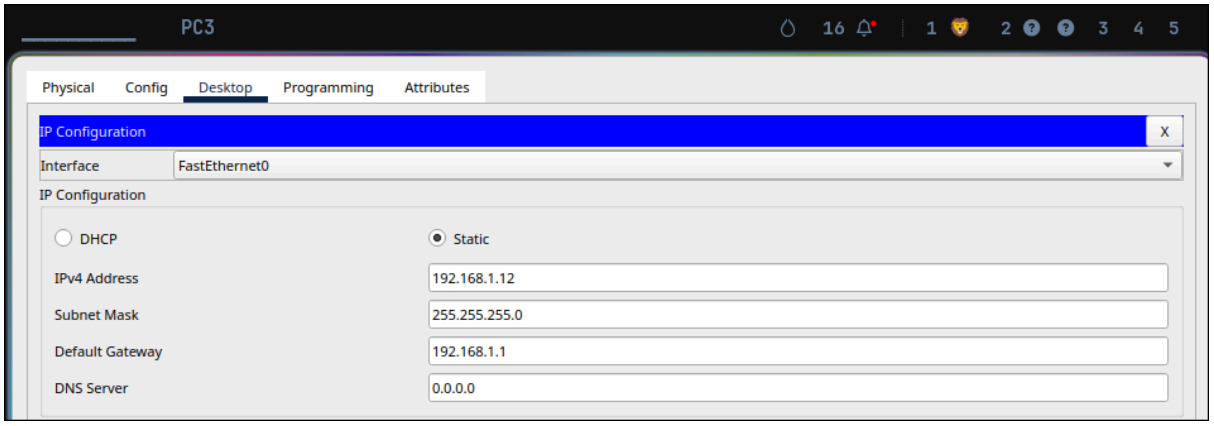
Subnet mask for all pc: 255.255.255.0

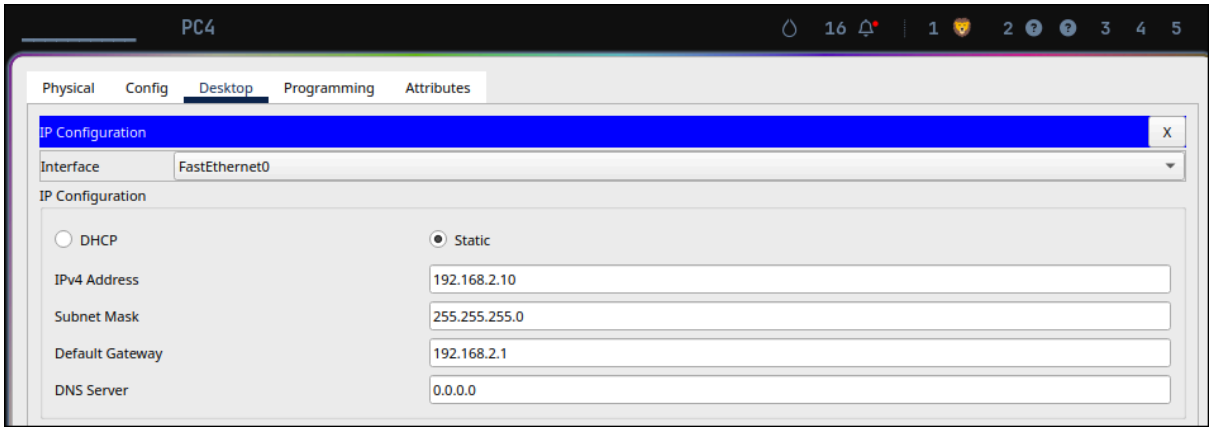
PC1 IP: 192.168.1.10/24
PC2 IP: 192.168.1.11/24
PC3 IP: 192.168.1.12/24
PC4 IP: 192.168.2.10/24
PC5 IP: 192.168.2.11/24
PC6 IP: 192.168.3.10/24

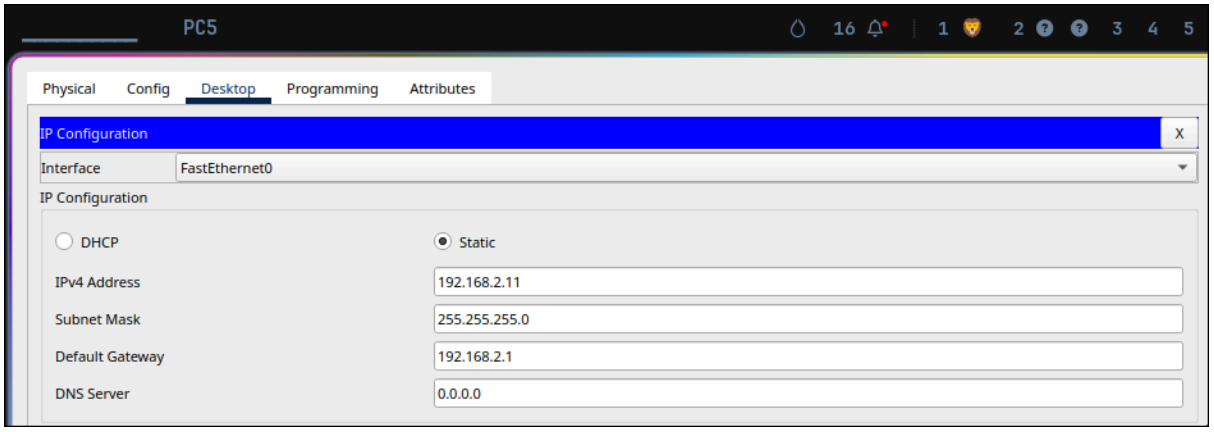
PC1,PC2,PC3 default gateway: 192.168.1.1
PC4,PC5 default gateway: 192.168.2.1
PC6 default gateway: 192.168.3.1

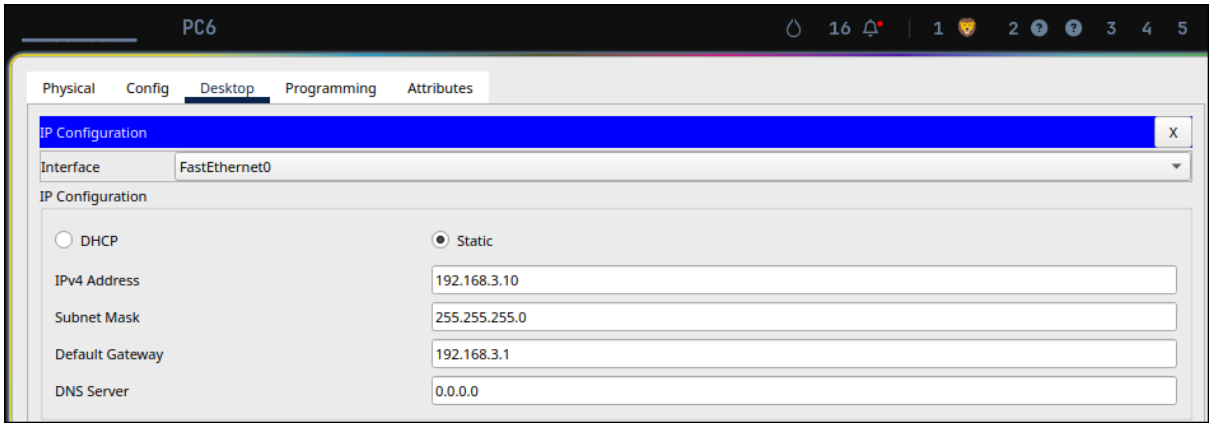





PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0


PC5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0


PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.3.1
DNS Server
0.0.0.0

Each PC requires an IP address, a subnet mask, and a default gateway to communicate.

- **IP Addressing:** Standard Class C private IP addresses (**`192.168.x.x`**) were used for the local networks . A **`/24`** subnet mask (**`255.255.255.0`**) was applied to all PCs, allowing up to 254 usable host addresses per local network.
- **Default Gateway:** For a PC to communicate outside its own LAN, it must know the address of its local router interface. Therefore, the default gateway for PC1, PC2, and PC3 was set to Router1's local IP (**`192.168.1.1`**). PC4 and PC5 were pointed to Router2 (**`192.168.2.1`**) , and PC6 was pointed to Router3 (**`192.168.3.1`**).

ROUTER configuration:

Subnet mask for all router: 255.255.255.252

Router1
gig0/0: 10.0.0.1/30
gig0/1: 10.0.0.10/30
gig0/2: 192.168.1.1/24

Router2
gig0/0: 10.0.0.2/30
gig0/1: 10.0.0.5/30
gig0/2: 192.168.2.1/24

Router3
gig0/0: 10.0.0.6/30
gig0/1: 10.0.0.9/30
gig0/2: 192.168.3.1/24

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 10.0.0.10 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#int g0/2
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.9
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

(router 1)

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#int g0/2
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.6
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

(route 2)

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 10.0.0.9 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#int g0/2
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.10
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.5
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

(router 3)

By default, Cisco router interfaces are administratively down. Using the Command Line Interface (CLI), each active **`GigabitEthernet`** port was configured with an IP address and brought online using the **`no shutdown`** command .

- **WAN Subnetting Strategy:** Instead of a **`/24`** mask, the router-to-router links were configured with a **`/30`** subnet mask (**`255.255.255.252`**). This is a network design best practice because a **`/30`** subnet provides exactly two usable IP addresses, perfectly matching the point-to-point connection between two routers and preventing IP address wastage.

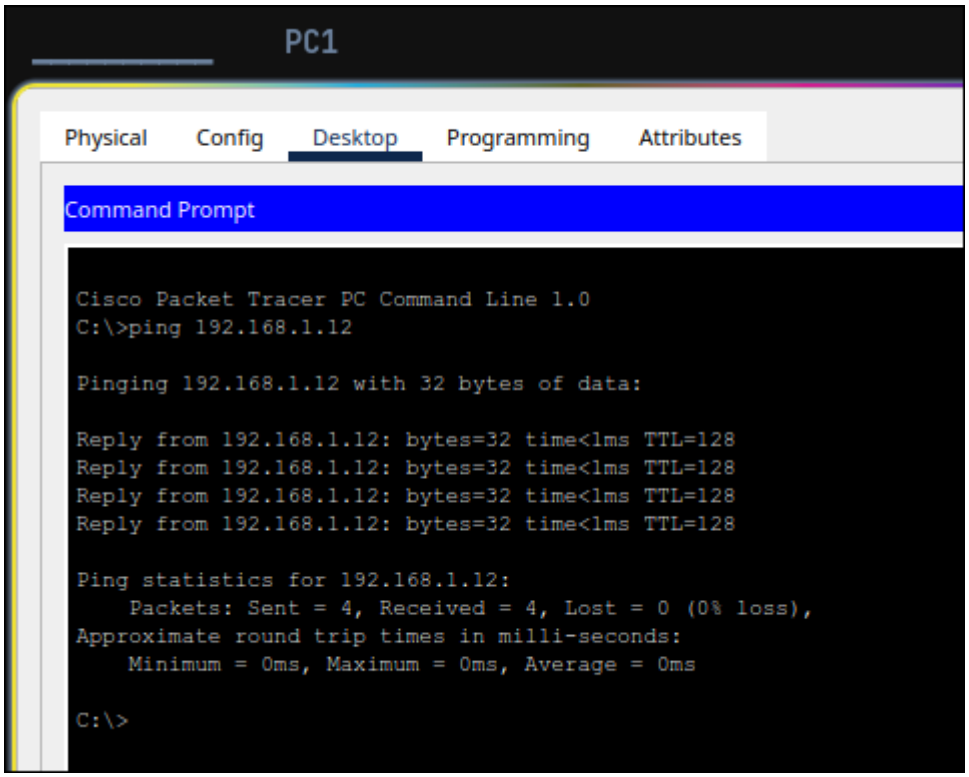
Because routers only know about networks physically plugged into them, static routing was required to allow cross-network communication. The **`ip route [destination_network] [subnet_mask] [next_hop_ip]`** command was utilized.

- For example, Router1 was configured with the command **`ip route 192.168.2.0 255.255.255.0 10.0.0.2`**. This manually dictates that any packet intended for

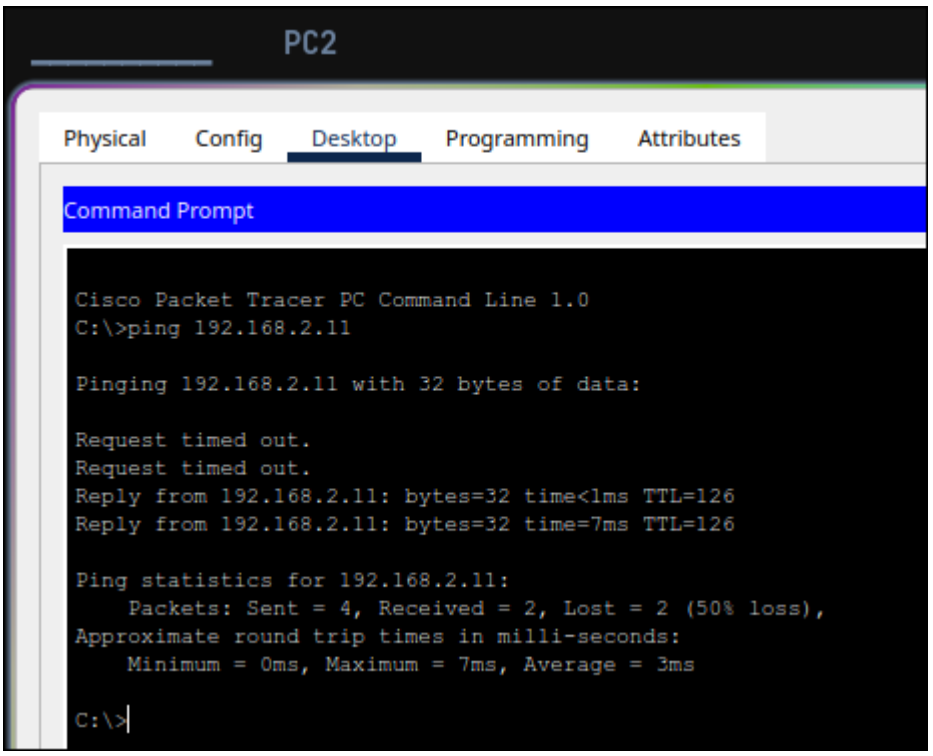
the **`192.168.2.x`** network (where PC4 and PC5 reside) must be forwarded out to Router2's interface at **`10.0.0.2`**.

PART C: VERIFICATION

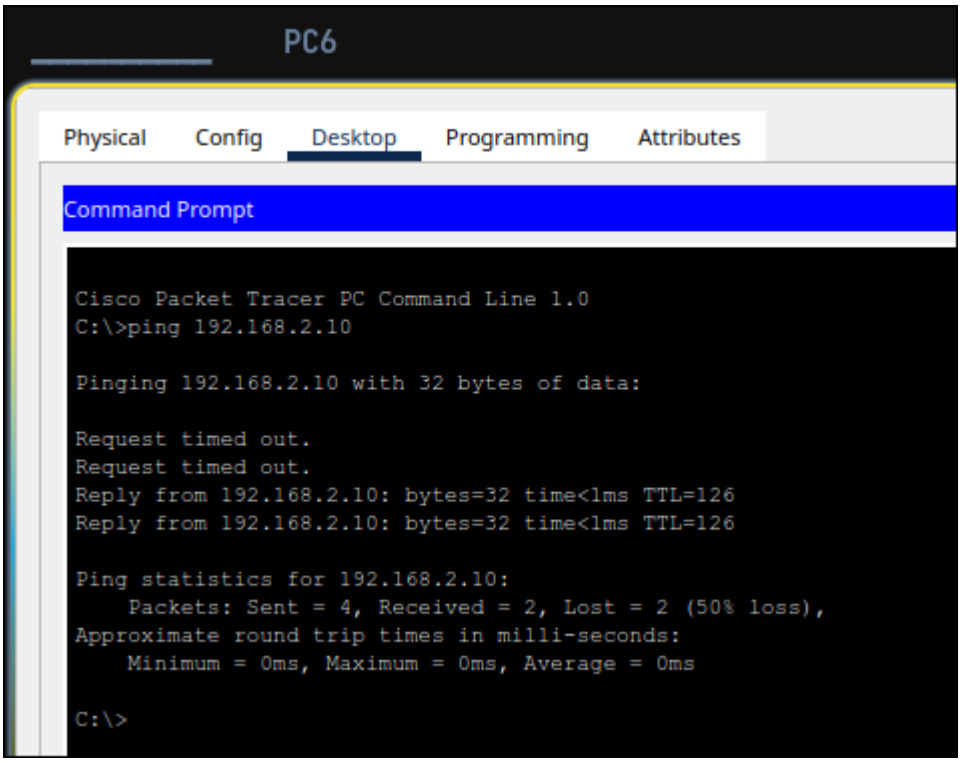
1. Ping:
    a. PC1 -> PC3



    b. PC2 -> PC5



c. PC6 -> PC4

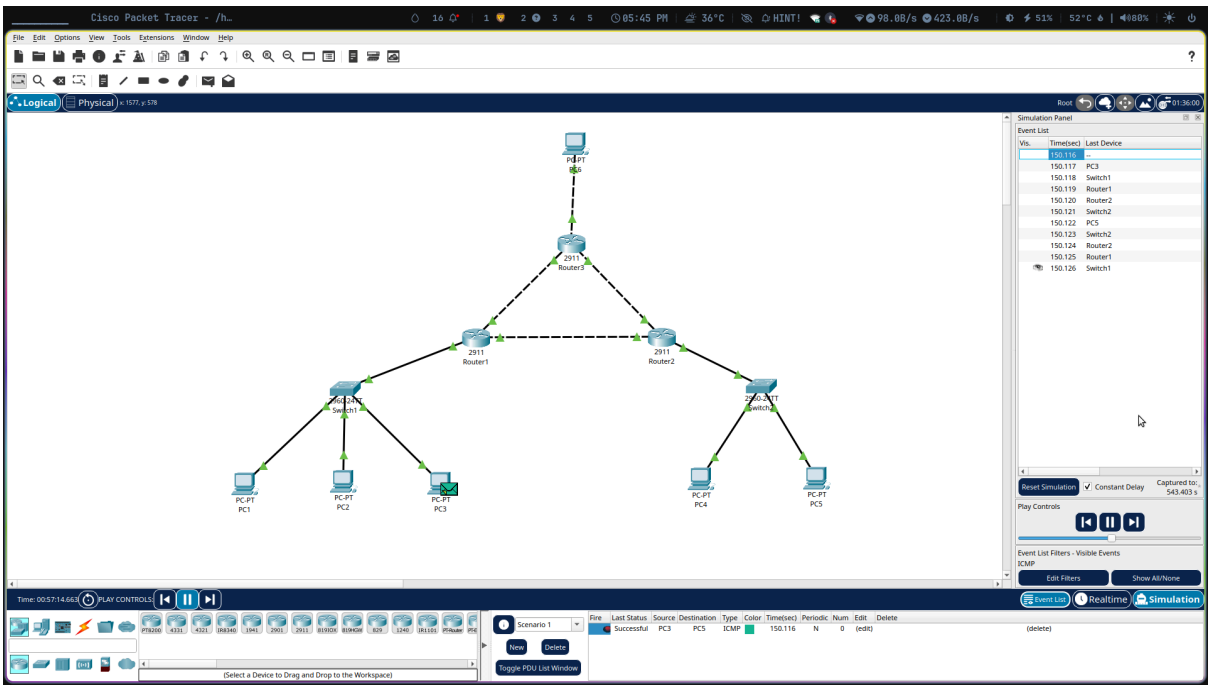


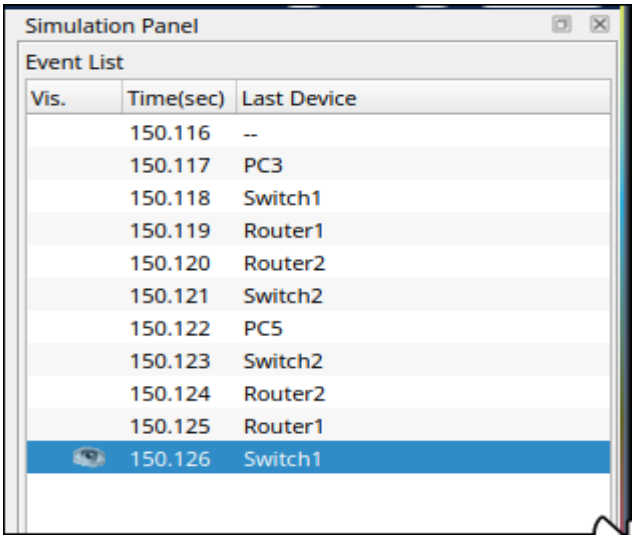
The **`ping`** command relies on the Internet Control Message Protocol (ICMP) to verify end-to-end connectivity.

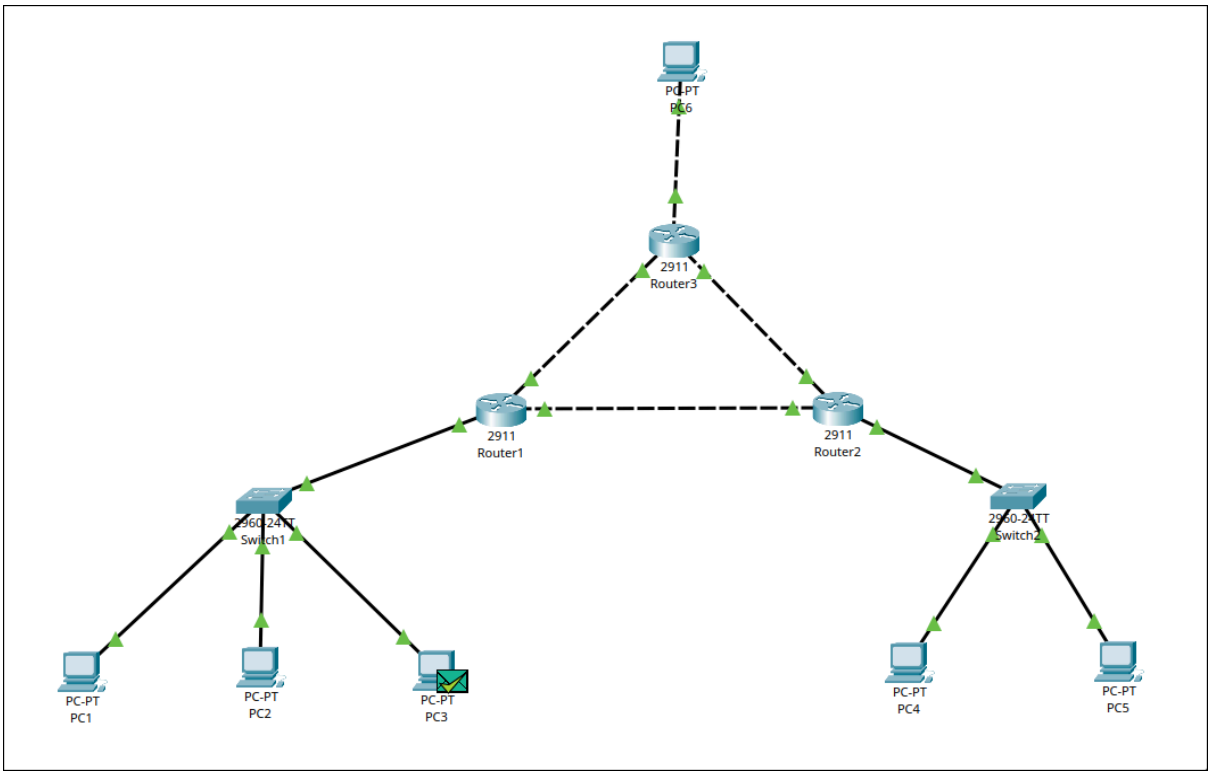
- **Local Routing:** Pinging PC3 from PC1 (**`ping 192.168.1.12`**) was immediately successful with **`<1ms`** response times because the traffic only had to traverse the local switch, never hitting the router .

- **Cross-Network Routing & ARP:** When pinging across the WAN (e.g., PC2 to PC5 using **`ping 192.168.2.11`**), the first two ICMP packets resulted in a "Request timed out" error . This is standard network behavior. The timeout occurs because the devices are executing an Address Resolution Protocol (ARP) broadcast to map the required IP addresses to their physical MAC addresses along the newly established static routes. Once the ARP tables were updated, the subsequent packets successfully reached the destination and returned .

2. Use simulation mode to check packet flow.







(PC3 to PC5)

Packet Tracer's Simulation Mode provides visual confirmation that the static routing tables are functioning exactly as designed.

- In the recorded simulation event for an ICMP packet sent from PC3 to PC5 , the Event List accurately logs the packet's physical hop-by-hop traversal .

- The log proves the packet successfully moved from the source (PC3), up to the local gateway (Switch1 -> Router1), across the WAN link dictated by the static route (Router1 -> Router2), and down to the target LAN (Router2 -> Switch2 -> PC5) before successfully returning along the same path . This validates the entire network topology and routing configuration.